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Artificial neural network chemistry solving for high-pressure hydrogen–air combustion

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ABSTRACT

The simulation of the time evolution of chemical species in hydrogen flames by artificial neural networks (ANN) is discussed for operating conditions representative of aeronautical combustion chambers, e.g., above the atmospheric pressure, in the presence of recirculating burnt gases and secondary air-dilution. This machine learning strategy aims at replacing the integration of ordinary differential equations for the computation of chemical source terms, which can be costly, particularly at high-pressure conditions. The training database is obtained from a population of stochastic particles undergoing turbulent micro-mixing and chemical reactions. The ANN performance is then validated a posteriori on the same micro-mixing canonical problem, replacing the chemistry integration with the ANN regression. Various operating conditions are addressed in terms of global fuel/air ratio, dilution by cooling air, and burnt gases, at 1 bar and 20 bars pressure levels. Two chemical schemes are employed: a 9-species mechanism for H_2 chemistry and a 15-species mechanism containing a reduced NOx sub-mechanism. Various neural network training methodologies from the literature are tested and compared. It is found that the most effective strategies lie in data addition, underlying the importance of the database in ANN training procedures. Furthermore, existing methodologies fail when accounting for NOx chemistry. An alternative network architecture and training method are therefore proposed and tested. The accurate monitoring of atomic mass conservation is also addressed.

1. Introduction

Hydrogen is one of the fuels retained to tackle the major challenge of decarbonizing air transportation, as its combustion does not contribute to direct CO_2 emissions [1–7]. Nevertheless, replacing conventional fuels with H_2 in aeronautical combustion chambers poses several technical difficulties. First, H_2 particular combustion properties (enhanced flashback and autoignition risks, thermo-diffusive instabilities, high flame speed [8]) make it challenging for aircraft engine architecture. Secondly, despite resolving the problem of soot emissions, H_2 combustion produces nitrogen oxides (NOx), which are regulated due to their contribution to the greenhouse effect, as well as the danger they represent for human health and the environment. While H_2 combustion with air involves a few chemical species and remains computationally affordable, adding NOx sub-mechanism complexifies the chemistry. Furthermore, in an aeronautical combustion chamber, H_2 burning requires high-pressure operating conditions, which steepened the gradients of the scalars to be solved.

In contexts where the design of optimized combustion systems relies on numerical simulation, the need for acceleration makes neural networks a compelling solution for replacing costly chemical integration. Alternative approaches for reducing computational expenses associated with chemical integration include chemical scheme reduction [9] and tabulation [10]. However, eliminating species and reaction steps generally results in reduced accuracy. Tabulated chemistry, which involves storing precomputed chemical values in large tables for retrieval during simulations, often presents memory usage challenges due to the database size and dimensions required to maintain precision.

Introducing neural networks is a third approach that addresses both of these shortcomings. Because of the attractiveness of deep learning applications in combustion over the last few years, multiple neural network methodologies have been proposed recently in the literature. Among others, convolutional neural networks (CNN) and artificial neural networks (ANN) are the main frameworks for chemistry prediction.

CNN is a neural network architecture used widely in image classification and recognition. A CNN framework was recently proposed for hydrogen combustion to replace a whole unsteady RANS solver, and predicted with accuracy velocity fields and species mass fractions, with a factor 100 acceleration compared to the CFD solver [11]. As

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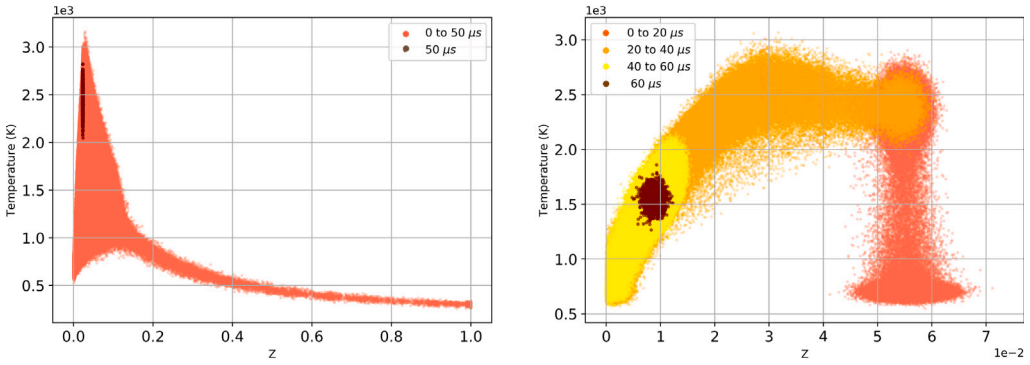


Fig. 1. Scatter plot of the time history of the temperature for all stochastic particles. Left: 2,496,751 data points of Case 2 (Table 3). Right: 3,406,850 data points of Case 5 (Table 4).

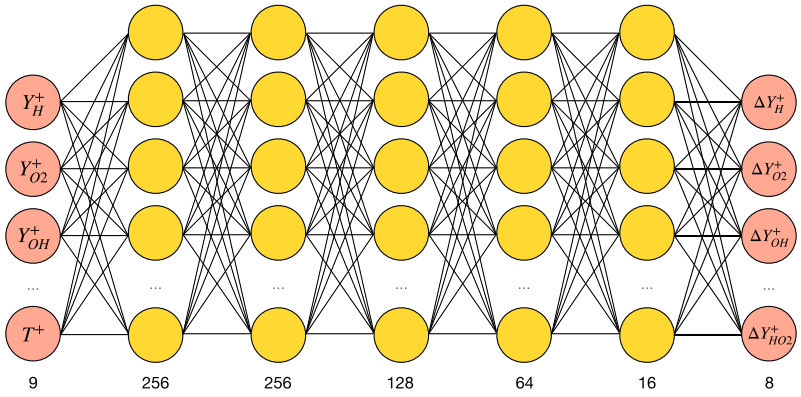


Fig. 2. MLP configuration for the 9-species mechanism.

might be expected, such an extreme approach is dependent of the training geometry, making the repeatability on other domains a limiting factor. However, it does show great potential for the development of fast-performing and accurate reduced-order models at industrial scale.

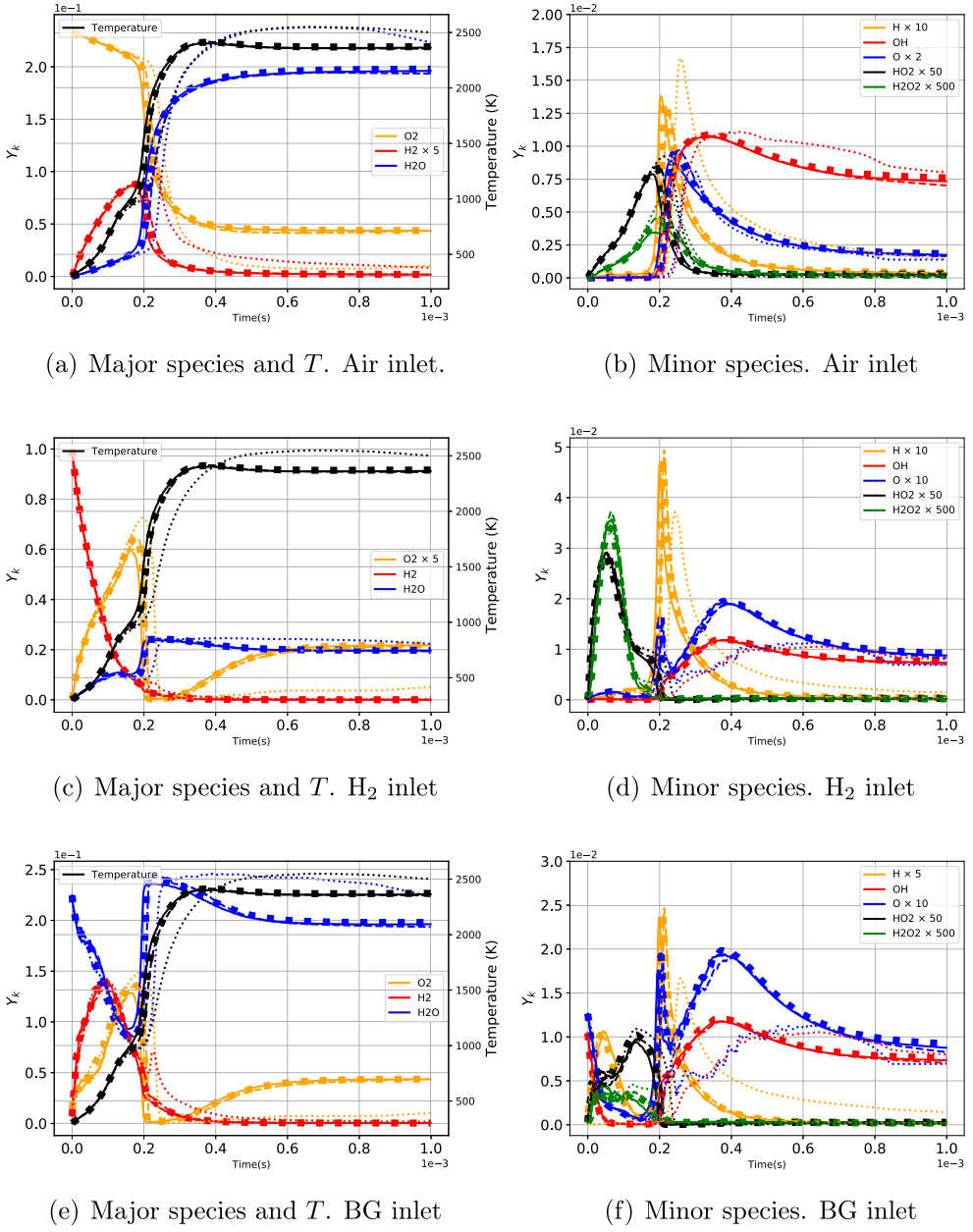
ANN, specifically multi-layer perceptron (MLP), eliminates geometric constraints as it solely performs in the composition space for predicting chemical quantities evolution in time when coupled with a CFD solver. One of the early attempts to introduce ANN in the combustion field was in 1998 by Blasco et al. [12,13], who trained an ANN to replace a stiff ODE solver for solving in time the integration of the chemical source terms. It can be seen as a nonlinear regression function with multiple coefficients that are adjusted during a training phase, that can handle multiple input and output data. In the case of chemical source terms prediction, input and output are respectively a state vector of chemical quantities and its increment over a fixed time step. This method is meant to be from 4 to 300 times faster than direct integration, as documented in the literature [14], and allow a time-step selection based on the flow constraints (CFL or Fourier numbers) instead of the chemistry stiffness in numerical simulations. However, artificial neural networks are known to perform poorly if encountering input data outside the range of the training dataset. For this reason, a slight output error can be amplified through a simulation and result in a significant divergence from the reference results at the end of the computed physical time.

Among all the proposed approaches from the literature, some are straightforward and others more sophisticated, depending on the chemical mechanism of reference and test case. They vary mostly in the number of MLP used (a single network for predicting all chemical species [14–17], one per species [18–21], or one for a subset of species [22]). The partitioning of the composition space (e.g. versus fuel equivalence ratio, temperature level, radical pools, etc.) is also a key ingredient. The preprocessing strategy and other parameters also vary, such as the data normalization technique and hyperparameters choice (loss functions, activation function, number of layers, and neurons per layer), but seem less critical.

Recently, Vijayarangan et al. [23] proposed a novel framework in the chemistry domain, using an auto-encoder coupled with a neural ordinary differential equation (NODE), instead of the classical dense layer MLP approach, and assessed it on a 1 atm hydrogen premixed flame. Üstün al. [24] have discussed a machine learning framework based on Gaussian process regression and autoregression. The benefits of this framework, compared to the standard MLP, are its partial interpretability and uncertainties quantification. However, it appears to be slower and has demonstrated greater efficacity on small datasets.

Several studies have focused on incorporating mass conservation and physical laws into ANN predictions [25,26]. Direct inclusion of physical restrictions into the training phase is one of the methods employed, as using a *softmax* activation function on the output layer, which bounds the prediction between 0 and 1 and ensures the sum of the outputs to be unity [27,28]. Another approach is the *a posteriori* correction of species mass fractions [29], ensuring atomic balance, which is of importance for computing the right temperature in CFD codes through the relation between sensible energy, heat capacities, and temperature.

This work aims to evaluate various ANN methodologies on a large dataset, ultimately developing a model to predict chemical source terms for H_2-air combustion, incorporating the NOx sub-mechanism. The study focuses on operating conditions relevant to aeronautical combustion chambers, at realistic pressures, and is applicable to multi-regime combustion and turbulent flows. Operating above atmospheric pressure affects the distributions of chemical time scales associated with fuel oxidation, the intermediate species radical pool, and NOx emissions. It is shown that this necessitates adjustments to previously developed ANN training procedures.



(a) Major species and T. Air inlet.

(b) Minor species. Air inlet

(c) Major species and T. H_2 inlet

(d) Minor species. H_2 inlet

(e) Major species and T. BG inlet

(f) Minor species. BG inlet

Fig. 3. Time-averaged evolution over particles per inlet (BG: Burnt Gases). Case 1 of Table 1, 1 bar, global equivalence ratio $\phi = 0.8$. ■ ■ : Sandiego mech. ——: Method A – – –: Method B ·····: Method C.

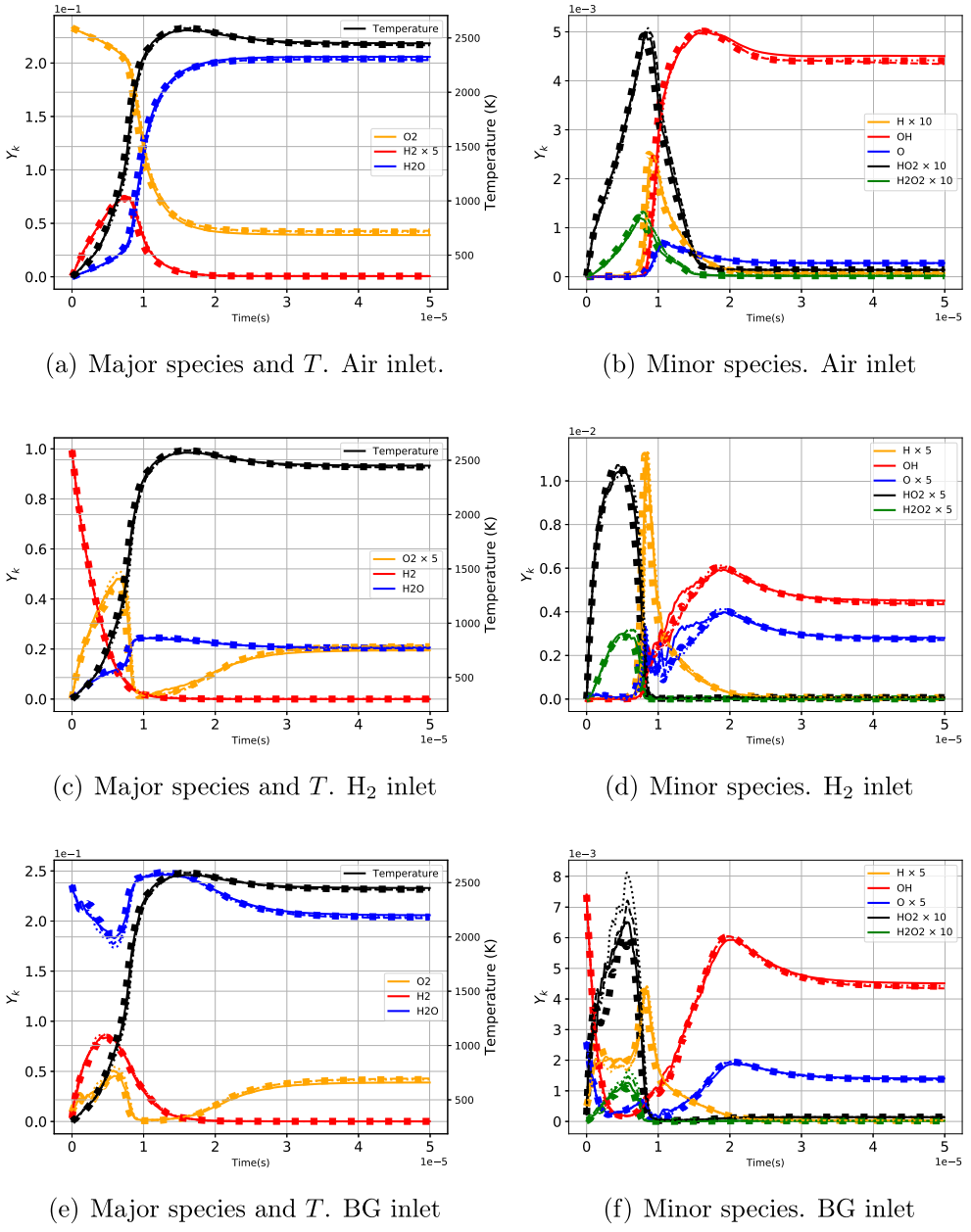
To ensure that ANNs not only reproduce the chemical source response at specific points (*a priori tests*) but also accurately capture the dynamic evolution of combustion from fresh to burnt gases, we report a *posteriori* tests where the ANN fully drives the simulation. In these tests, any error accumulation becomes immediately apparent through deviations from the physical response. Additionally, a training process, based on stochastic micro-mixing, ensures that the full range of realizable composition space conditions is covered by the ANN. This makes the ANN robust to the dynamic variations present in flows.

The coupling with flow transport introduces additional challenges, which were addressed in previous works without requiring significant modifications to the training procedure or ANN architecture [15,20,30].

In cases where the time history exhibits strong dynamical effects, such as those observed in soot chemistry, Long Short-Term Memory (LSTM) networks have been applied [31]. However, such complexity has proven unnecessary for hydrogen chemistry, which remains relatively simple. The most sensitive aspect is the training process's organization, which is this paper's focus.

2. Operating conditions and detailed chemistry

The operating conditions investigated here are designed to approximate those of a realistic aeronautical combustion chamber. The setup includes a primary inlet of hot, pressurized air at 700 K from upstream compressors, a secondary inlet of H_2 at 300 K, and a tertiary inlet of burnt gases at stoichiometric thermal equilibrium, added to simulate recirculating burnt gases in a swirling flow. These gases mix and react following a procedure outlined below, with the hot gases ensuring ignition of the mixture. The goal is to develop a generalized network methodology adaptable to the most demanding scenarios, specifically multi-regime combustion, across various global equivalence ratios. The combustion chamber is designed to operate at 20 bars. A summary of the four test cases is provided in Table 1.



(a) Major species and T. Air inlet.

(b) Minor species. Air inlet

(c) Major species and T. H_2 inlet

(d) Minor species. H_2 inlet

(e) Major species and T. BG inlet

(f) Minor species. BG inlet

Fig. 4. Time-averaged evolution of over particles per inlet (BG: Burnt Gases). Case 2 of Table 1, 20 bars, global equivalence ratio $\phi = 0.8$. ■ ■ : Sandiego mech. ——: Method A – – – Method B ·····: Method C.

Table 1
Initial conditions for test cases 1 to 4, mimicking diffusion/partially premixed flames.

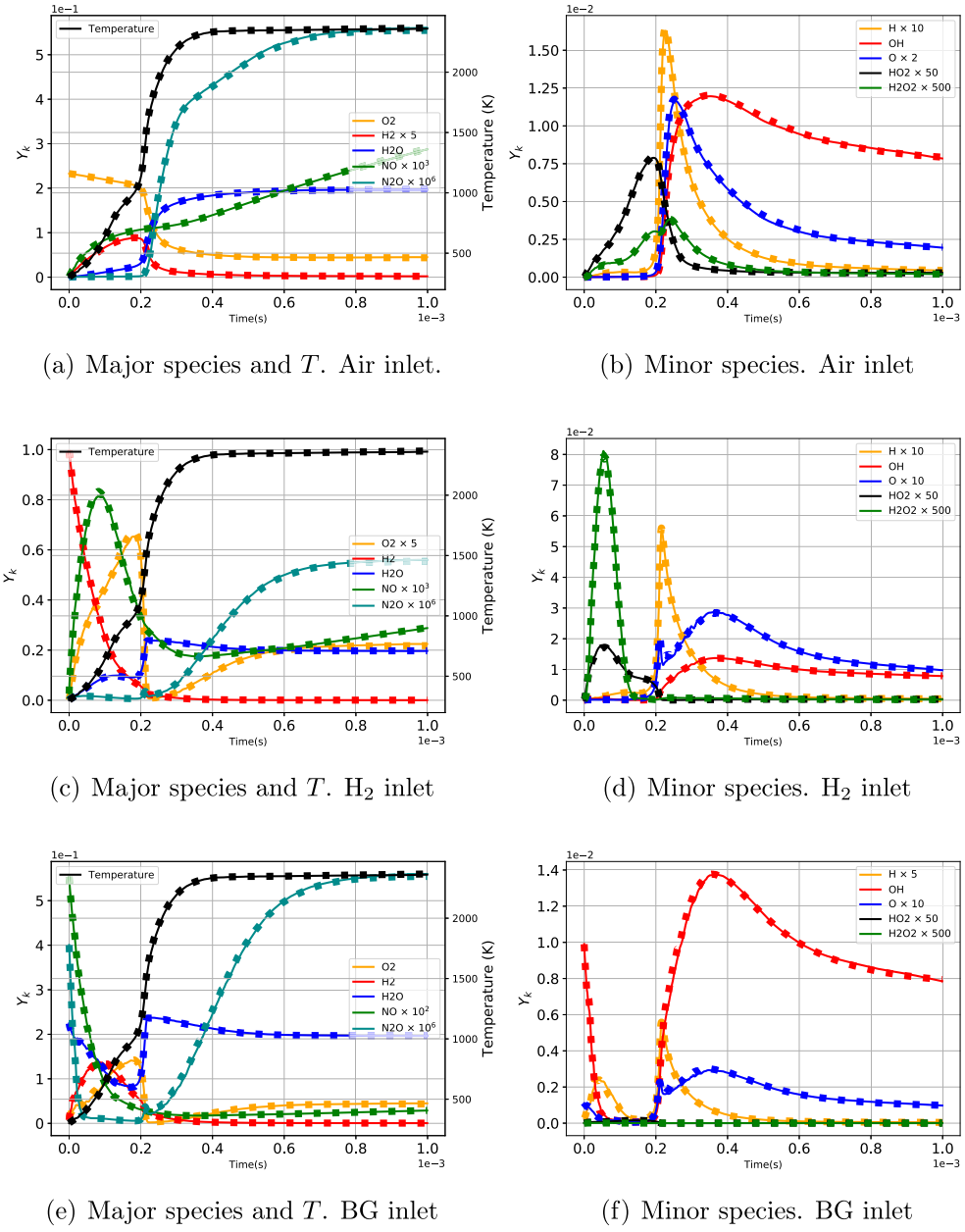
	Pressure	Global ϕ	Temperature (K)		Dilution
			H_2	Air	Burned gas
Case 1	1 bar	0.8	300	700	2561 K, 2.1%
Case 2	20 bars	0.8	300	700	2678 K, 5.9%
Case 3	1 bar	0.4	300	700	2561 K, 2.1%
Case 4	20 bars	0.4	300	700	2678 K, 5.9%

Table 2
Initial conditions for test case 5, mimicking a RQL burner.

	Pressure	Global ϕ	Delay of injection		
			H_2+Air ($\phi = 2$)	Air 1	Air 2
Case 5	20 bar	0.4	t = 0 μs	t = 20 μs	t = 40 μs

A fifth case, even more specific to aircraft combustion chamber, is added with a global equivalence ratio of $\phi = 0.4$. It is inspired by the RQL (Rich-Burn, Quick-Mix, Lean-Burn) burner, adapted to hydrogen combustion. It consists of the fully burnt products of a rich premixed mixture at $\phi = 2$ and 700 K, followed by a quick addition of air at 700 K staged in time with a delay of 20 μs, to burn off the remaining hydrogen with a globally lean mixture. At the end of the burning process (t = 40 μs), another addition of air is performed, again at 700 K, in order to cool down the combustion products at a temperature that would be representative of their transfer to the turbine.

The chemical schemes used for this study are the well-known hydrogen–air Sandiego 9-species mechanism from 2016 [32], and a 15-species reduced mechanism from the CRECK, accounting for nitrogen chemistry [33]. The methodology discussed is designed to have low sensitivity to the chosen H_2-air chemical mechanism.



(a) Major species and T. Air inlet.

(b) Minor species. Air inlet

(c) Major species and T. H_2 inlet

(d) Minor species. H_2 inlet

(e) Major species and T. BG inlet

(f) Minor species. BG inlet

Fig. 5. Time-averaged evolution over particles per inlet (BG: Burnt Gases). Case 1 of Table 1, 1 bar, global equivalence ratio $\phi = 0.8$. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B.

3. Training database

Several parameters play a role in the construction of an ANN model. First, the network architecture and hyperparameters, and secondly, the training database.

3.1. Methodology for exploring H_2 chemical composition space

The training database is generated through a 0D canonical turbulent micro-mixing problem, which was implemented in a previous work focusing on chemistry reduction [34,35], and validated for neural network training in several others [14,15,17].

A set of stochastic particles are initialized for each inlet to represent the system's initial state, and will then evolve through mixing and combustion (chemical reactions) at each time-step Δt. The number of particles is chosen proportionally to the mass flow rate corresponding to the targeted global equivalence ratio. As an example, the initialization for case 1 (cf. Tables 1 and 3) consists of three groups of particles: 1152 air, 27 H_2, and 25 burnt gas, each one set to their respective temperature.

Starting from the initial state, each iteration of the micro-mixing problem involves mixing the mass fractions and sensible enthalpy of all stochastic particles, followed by the chemical reaction, as expressed for any particle p:

$$\frac{dY_k^p}{dt} = MIX_k^p(\tau_T) + \dot{\omega}_k^p, \tag{1}$$

$$\frac{dh_s^p}{dt} = MIX_{h_s}^p(\tau_T) + \dot{\omega}_{h_s}^p. \tag{2}$$

The micro-mixing stage denoted MIX in Eqs. (1) and (2) is performed using the Euclidean Minimum Spanning Trees (EMST) micro-mixing model [36], and the reaction one ($\dot{\omega}_k^p$ and $\dot{\omega}_{h_s}^p$) is carried out

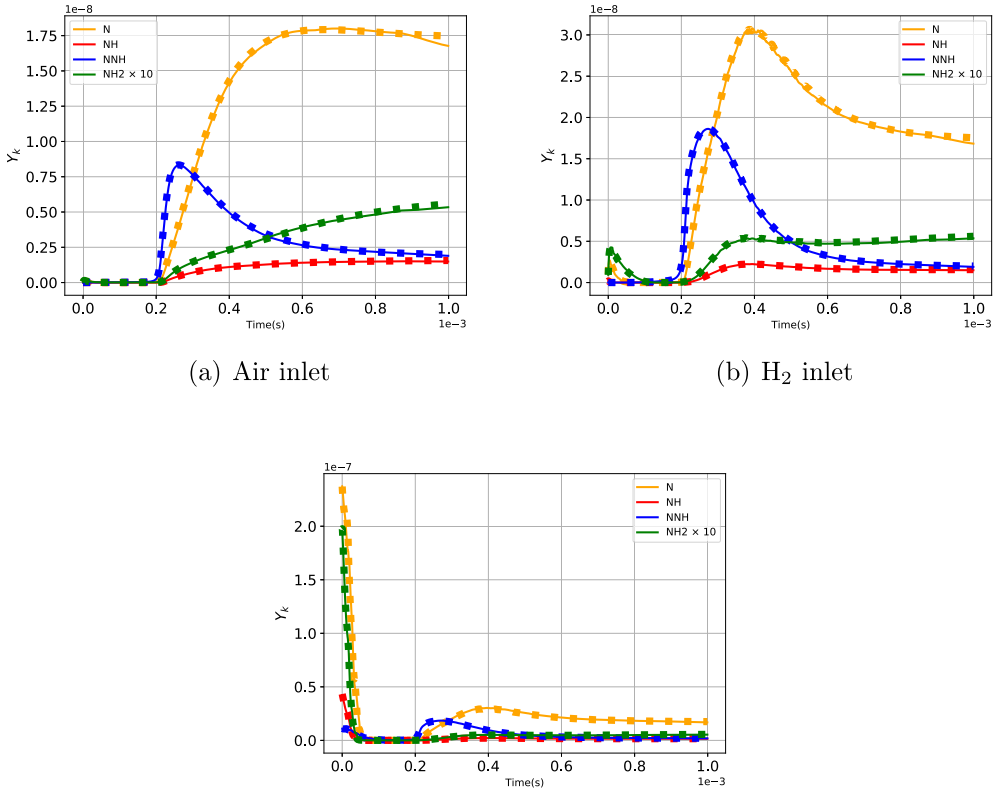


Fig. 6. Time-averaged evolution of NOx over particles per inlet (BG: Burnt Gases). Case 1 of Table 1, 1 bar, global equivalence ratio $\phi = 0.8$. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B.

Table 3
Input parameters for test cases 1 to 4.

	τ_T (s)	Δt (s)	Number of particles		
			H_2	Air	Burnt gas
Case 1	1×10^{-4}	5×10^{-7}	27	1152	25
Case 2	0.5×10^{-5}	2.5×10^{-8}	27	1152	70
Case 3	1×10^{-4}	5×10^{-7}	26	2214	25
Case 4	0.5×10^{-5}	2.5×10^{-8}	26	2214	70

Table 4
Input parameters for test case 5.

	τ_T (s)	Δt (s)	Number of particles		
			H_2+Air ($\phi = 2$)	Air 1	Air 2
Case 5	0.5×10^{-5}	2.5×10^{-8}	225	850	340

by Cantera ideal gas constant pressure reactor which uses the stiff ODE solver CVODE [37]. τ_T is the characteristic micro-mixing time, representative of a turbulent eddy turnover time. Varying its value was found to be of little significance, provided it is not so small as to prevent ignition or so large as to inhibit the development of turbulence-chemistry interaction. The corresponding parameters for the cases in Table 1 are presented in Table 3.

The most notable effect of increased pressure is the linear rise in the maximum H_2O source term. Since the reaction proceeds 20 times faster at 20 bars, both the time step and simulated physical time were reduced by the same factor between the 1-bar and 20-bar test cases. This setup enables coverage of the composition space from initial mixing to chemical equilibrium, exhibiting specific behavior due to air dilution, as illustrated in Fig. 1.

As an example for the 9-species mechanism, each particle, containing a state vector of 8 species mass fractions (N_2, inert species, is not included) and temperature (Y_k, T) before and after the chemical integration step, forms a pair of input and output of the ANN training phase. Each test case will be carried out with both the semi-detailed H_2 9-species mechanism and the H_2 + NOx 15-species mechanism.

3.2. Database preprocessing and partitioning of the composition space

The ANN predicts all species mass fractions increments ΔY_k due to chemical reactions through a fixed time-step Δt. ANN-CNN models with time-step as an input have been proposed previously [31]. For coupling with CFD, it was found more practical to train the model using a small time step and perform subiterations (i.e., invoking the ANN multiple times) to match the CFD flow time step [20,30]. In any of the following training, the input data are log-transformed (after being clipped to 10^{-12}) to smooth out the data distribution and then rescaled with a mean-standard deviation normalization:

$$Y_k^+ = \frac{(\log Y_k - \langle \log Y_k \rangle)}{\sigma_{\log Y_k}}, \tag{3}$$

$$T^+ = \frac{(\log T - \langle \log T \rangle)}{\sigma_{\log T}}, \tag{4}$$

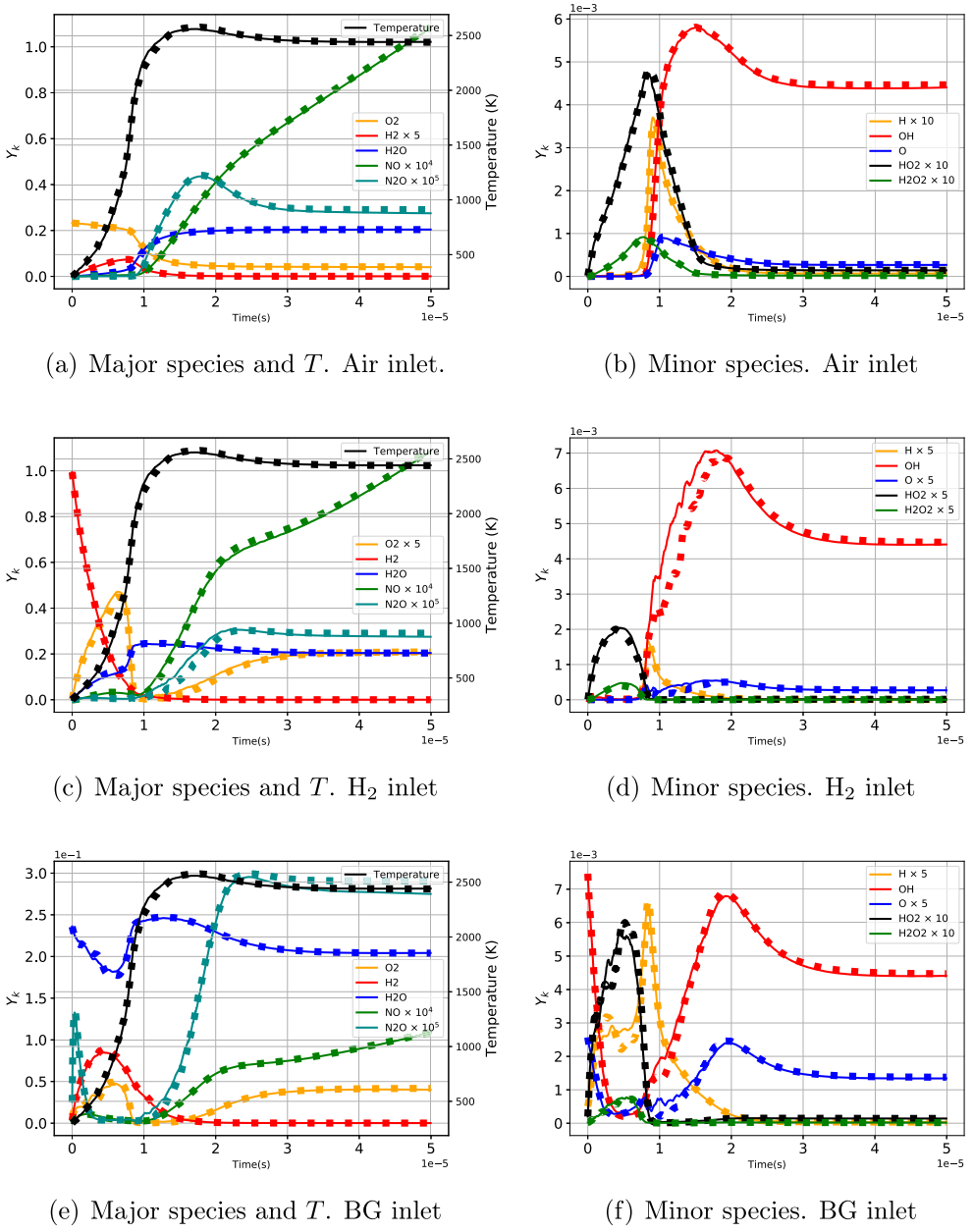
where $\langle \cdot \rangle$ denotes averaging over all particles.

The outputs for training and validation are constructed as the difference between pairs of mass fractions before and after reactions

$$\Delta Y_k = Y_k(t + dt) - Y_k(t), \tag{5}$$

before undergoing the same log-transformation and rescaling:

$$\Delta Y_k^+ = \frac{(\log \Delta Y_k - \langle \log \Delta Y_k \rangle)}{\sigma_{\log \Delta Y_k}}. \tag{6}$$



(a) Major species and T. Air inlet.

(b) Minor species. Air inlet

(c) Major species and T. H_2 inlet

(d) Minor species. H_2 inlet

(e) Major species and T. BG inlet

(f) Minor species. BG inlet

Fig. 7. Time-averaged evolution over particles per inlet (BG: Burnt Gases). Case 2 of Table 1, 20 bars, global equivalence ratio $\phi = 0.8$. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B.

After prediction, the outputs are de-normalized with the values of mean $\langle \log \Delta Y_k \rangle$ and standard deviations $\sigma_{\log \Delta Y_k}$ over all particles and all time-steps, stored from the training phase. Then, the values of the mass fractions are recovered as:

$$Y_k(t + \Delta t) = \exp(\log Y_k(t) + \Delta(\log Y_k)) \,, \tag{7}$$

with $\log \Delta Y_k$ being the de-normalized prediction of the ANN.

The temperature is not an output of the ANN and can be approximated after ANN chemistry prediction following the conservation of enthalpy[1]:

$$T(t + \Delta t) = T(t) + \frac{\sum_{k=1}^{nsp} \Delta Y_k \Delta h_k}{c_p} \,. \tag{8}$$

[1] This approach allows for introducing ANN also in compressible density coupled flow solvers.

In addition to the sampled database from the 0D micro-mixing problem, additional inputs are generated to enrich the dataset. A copy of the input database is created, with 0%–5% normal random noise applied to improve the network's robustness to slight variations in input data and to reduce overfitting, while ensuring atomic mass conservation [15]. Consequently, multiple inputs correspond to a single output.

The critical requirement is that the ANN must perform well during the ignition phase. This is particularly challenging due to the under-representation of ignition states in the dataset, stemming from the rapid ignition of H_2 combustion. Consequently, no strong correlations were observed between *a priori* and *a posteriori* validations in this study. While ANN training can yield excellent *a priori* results, any tiny inconsistency during ignition tends to cause significant deviations over the course of the entire *a posteriori* simulation. To address this, two methods (A and B) are tested to direct ANN training towards ignition states, with a third method (C) from the literature included for comparison, which does not rely on additional data.

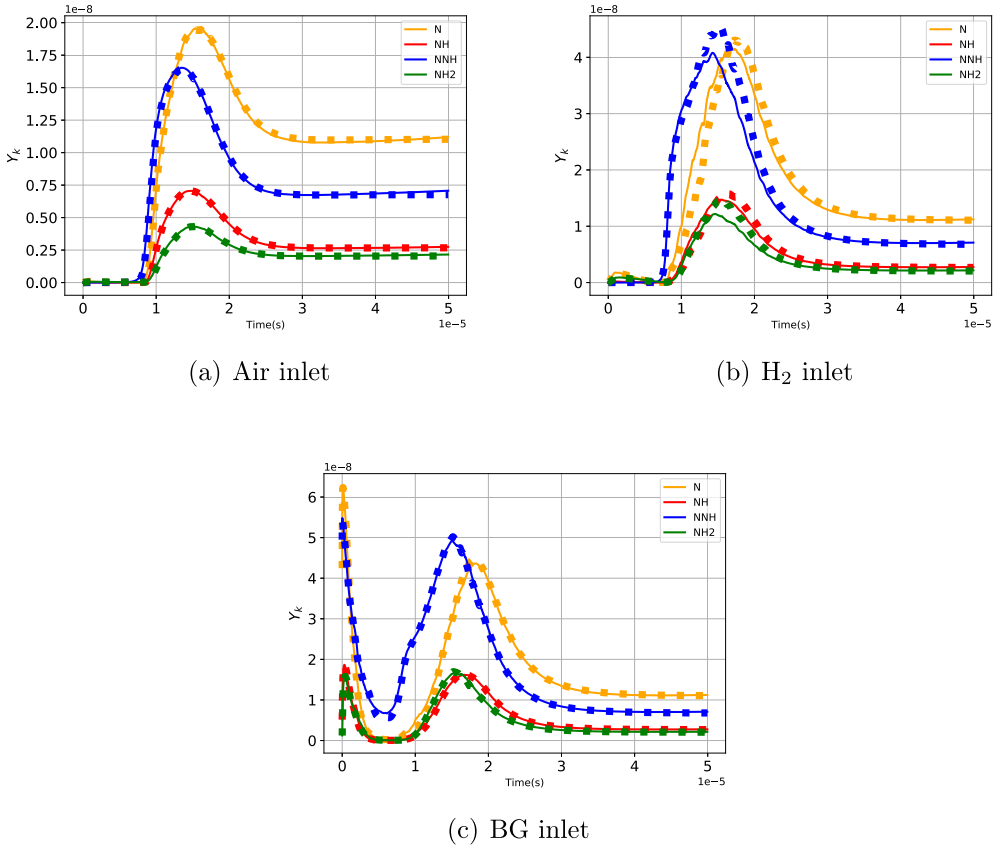


Fig. 8. Time-averaged evolution of NOx over particles per inlet (BG: Burnt Gases). Case 2 of Table 1, 20 bars, global equivalence ratio $\phi = 0.8$. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B.

3.2.1. Random data addition - Method A

The first approach is borrowed from [15]. H_2O_2 is one of markers of the fast ignition. To secure that the full range of H_2O_2 is seen during training (i.e. sufficient data points during the rapid ignition stage), a copy of the input database is generated, with all $Y_{H_2O_2}$ values replaced by random floating numbers with a uniform distribution between 0 and the maximum value of $Y_{H_2O_2}$. These additional state vectors are then reprocessed through the Cantera constant pressure reactor to get their output state vectors and added to the original dataset.

3.2.2. Sub-time-steps addition - Method B

A more physical approach would include sub-time steps from the constant pressure reactor calculation. The maximum source terms of H_2O_2, HO_2 and H_2O are computed, and an algorithm is coded (see Table 5) to retrieve a proportional number of sub-time-steps from 1% to 100% of the maximum source term of these three species. As for random data addition, the substates collected are then reprocessed through the Cantera constant pressure reactor to get their output state vectors.

For method A as well as for method B, the most rudimentary data clustering technique is applied. The cut is done arbitrarily between fresh gases and hot flow at 1000 K, meaning that when the state of a particle meets the cold flow condition (<1000 K), the prediction of its chemical state will be carried out by the cold flow ANN and *vice versa*. This specialization will allow each MLP to perform more efficiently in each region of the flow.

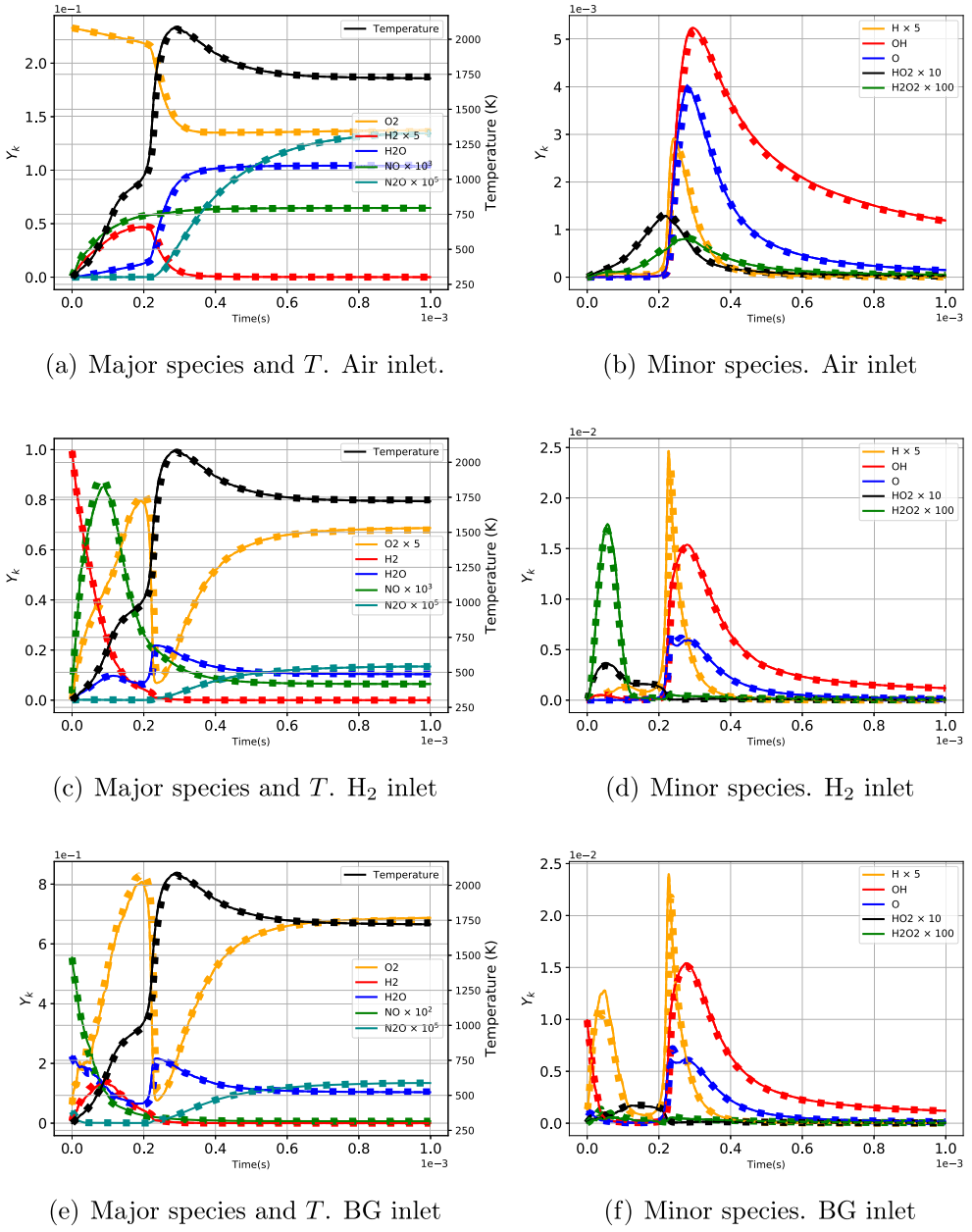
3.2.3. K-means clustering and LPCA - Method C

A different approach is also taken from [17]. No enrichment of the training database is needed, but the training and clustering processes are more demanding. The composition space is fractioned between 45 clusters using an unsupervised K-mean algorithm [38]. Each of these clusters represents a specific combustion stage from cold flow to equilibrium, allowing particles to have a similar distribution within each one. One MLP is trained for each partitioned composition space, meaning that a state particle would have its new composition computed by the MLP assigned to the cluster's centroid with the closest Euclidian distance. By allowing a specialization of each MLP for very specific parts of the composition space, each one is less prone to errors. This method was developed for syngas combustion, and all ANN parameters (normalization, MLP hyper-parameters, and size) are kept identical to those of this previous study [17] (i.e., the relevant quantities are not log-normalized). In addition, within each clustered composition space, the input data are also pre-processed through a Local Principal Component Analysis (LPCA) [39,40], which forms a new vector basis of the same dimension for describing the thermochemical states. This is intended to ease the ANN learning process.

4. Neural network architecture

4.1. 9-Species mechanism

The decision to use a single MLP to predict the complete vector of species versus mapping the input vector to individual species mass fractions–resulting in one MLP per species–remains an open question. The latter approach has been reported to yield higher precision under specific conditions and help prevent the ANN from converging to local minima during training [29], resulting in improved predictions for certain species. Ultimately, the choice involves a trade-off between computational cost and accuracy, as predicting nine species with the second method would require more operations by invoking nine separate MLPs for each particle, compared to a single MLP in the first method. However, an MLP with a single output would require fewer parameters (weights and biases) to achieve satisfactory accuracy. Since



(a) Major species and T. Air inlet.

(b) Minor species. Air inlet

(c) Major species and T. H_2 inlet

(d) Minor species. H_2 inlet

(e) Major species and T. BG inlet

(f) Minor species. BG inlet

Fig. 9. Time-averaged evolution over particles per inlet (BG: Burnt Gases). Case 3 of Table 1, 1 bar, global equivalence ratio ϕ = 0.4. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B.

Table 5
Algorithm to retrieve sub-steps states from constant pressure reactor for each time-step Δt from H_2O.

Require: $|\Delta Y_{H_2O}|_{max}$
  $\Delta Y_k \leftarrow Y_k(t+\Delta t) - Y_k(t)$
  if $|\Delta Y_{H_2O}| \geq 0.01|\Delta Y_{H_2O}|_{max}$ **then**
    $N \leftarrow int(\frac{|\Delta Y_{H_2O}|}{|\Delta_{H_2O}|_{max}} \times 100)$ ▷ N is the relative percentage of $|\Delta Y_k|$
    $n \leftarrow 0$
    while $n < N$ & $sim.time() < \Delta t$ **do** ▷ From time t to $t+\Delta t$
      $Y_k, T \leftarrow sim.step()$ ▷ Save each Y_k, T in database
      $n \leftarrow n+1$
    end while
  end if

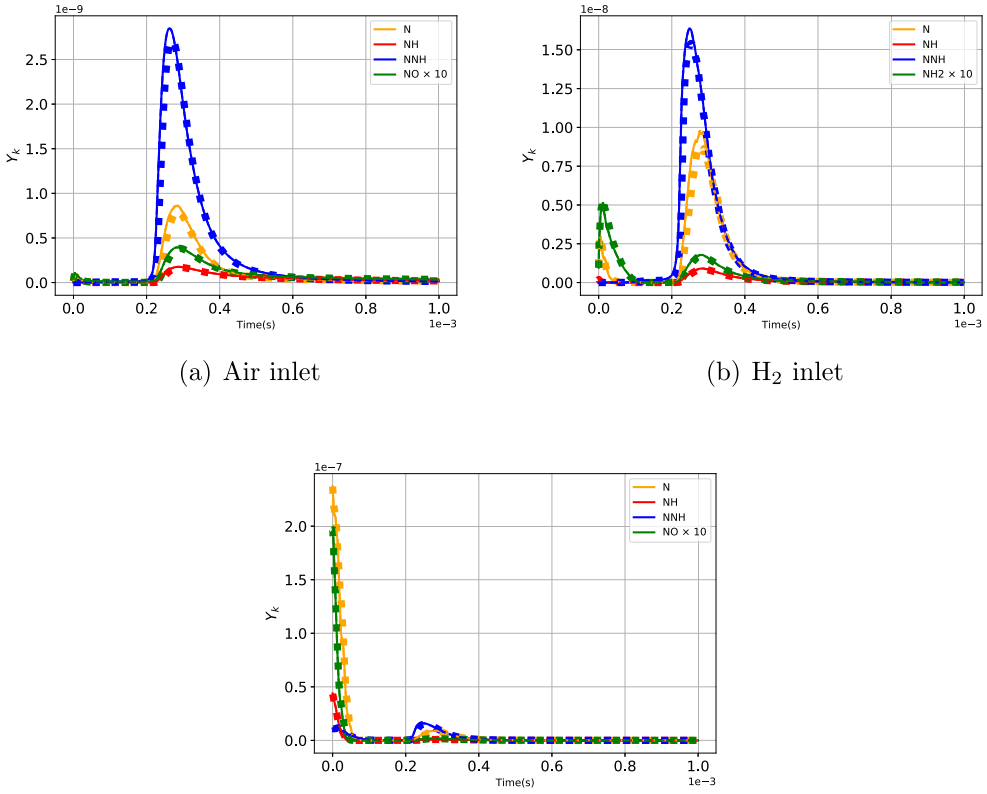


Fig. 10. Time-averaged evolution of NOx over particles per inlet (BG: Burnt Gases). Case 3 of Table 1, 1 bar, global equivalence ratio $\phi = 0.4$. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B.

the ANN cost is dependent on the number of parameters, the cost of an ANN call is influenced by this choice. In this study, it has been decided to predict all species using a single MLP.

The training phase utilizes the GPU-assisted Python TensorFlow library. The neural network architecture comprises five hidden dense layers, with 256, 256, 128, 64, and 16 neurons in each layer, respectively, all connected by the popular rectified linear unit (ReLU) activation function:

$$\mathrm{ReLU}(x) = \max(x, 0)\,. \tag{9}$$

The Adam optimizer [41] is used alongside the widely adopted Mean Squared Error (MSE) cost function to assess the network's output error. The dataset is divided into training, validation, and test subsets in proportions of 80%, 10%, and 10%. The training points and test subsets are first selected randomly. The remaining 10% of data compose the test subset specifically employed to evaluate the trained MLP a priori performance on unseen data from the training phase. Notice that every case is considered separately from others, as they do not share the same composition space response that would be seen in the turbulent flow.

Fig. 2 illustrates a schematic representation of the ANN architecture. Providing a definitive theoretical justification for the ANN structure in such a non-linear and stiff chemical system is beyond the scope of current knowledge. The network architecture is similar to that proposed in prior work [15,17]. ANNs with fewer intermediate layers have been tried without success. In the results discussed below, training achieves convergence after seeing 200 epochs without improvement of the mean-square error (about 1000 epochs for a single ANN to reach convergence).

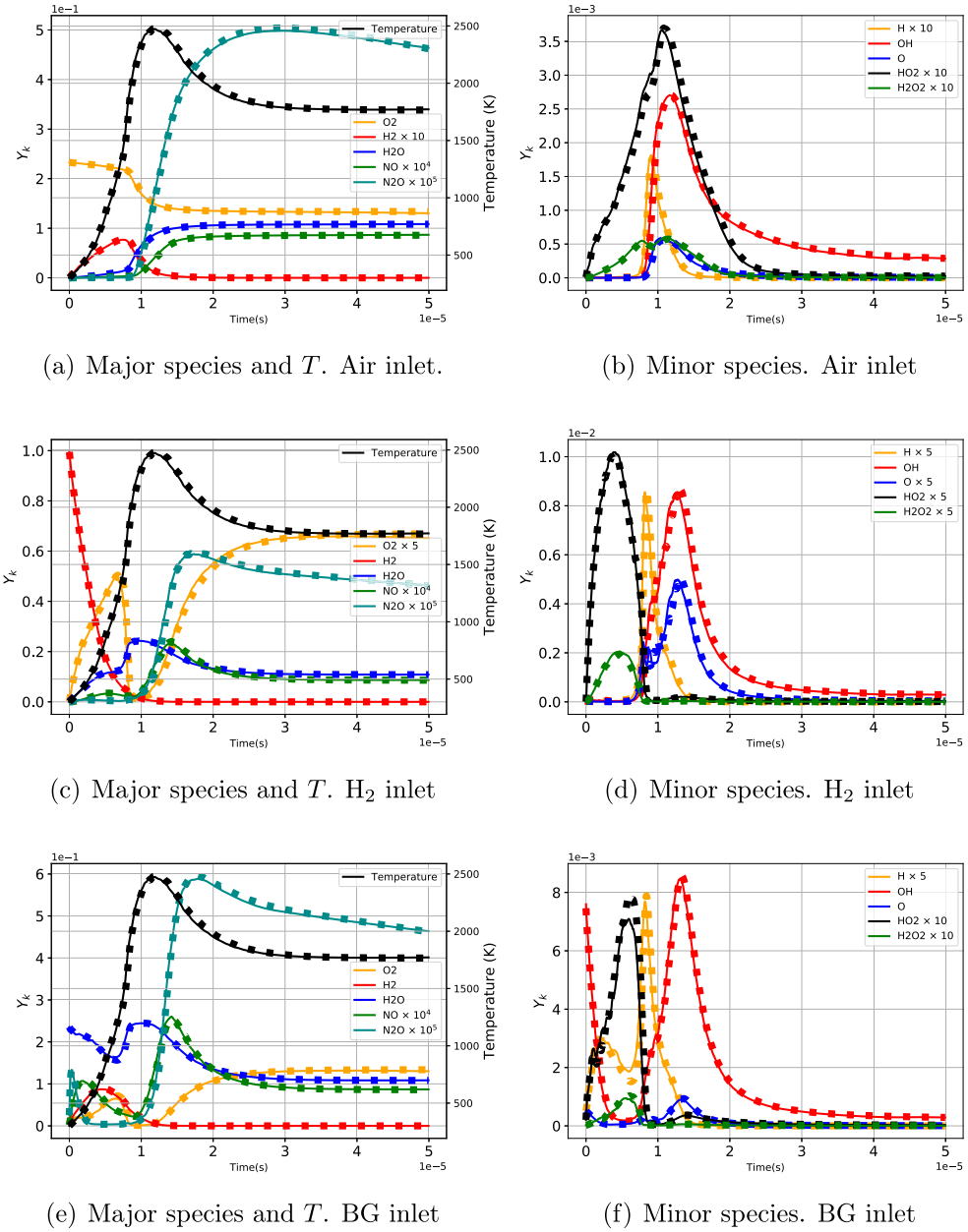
Training the 45 MLPs in method C is approximately 4.5 times more computationally expensive than training both the cold and hot MLPs in methods A and B, which takes about 3 h on a standard desktop.

4.2. Adding NOx species - 15-species mechanism

Applying the same method to the 15-species mechanism (which includes NOx species) resulted in poor outcomes. The MLP struggled to predict several species with significantly different time scales and behaviors, even when its architecture was expanded. Following the compromise philosophy presented in [22], which introduced MLPs that predict groups of species sharing similar correlations in large mechanisms, an additional network with the same architecture was integrated into the previous one. One MLP predicts the radical species H, OH, O, HO_2, H_2O_2, N, NH, NNH, and NH_2, while the other predicts the reactants and products O_2, H_2, H_2O, NO, and N_2O. This grouping of species with similar time scales and behaviors allows each MLP to make more accurate predictions, as they are tasked with predicting closely correlated data, albeit at an increased computational cost. Once again, N_2 serves as an input to the ANN but is not an output; it is calculated after each ANN prediction based on the conservation of nitrogen atoms.

5. Results

All methods presented were tested a posteriori for the San Diego 9-species mechanism on test cases 1 and 2 from Table 1, replacing the Cantera constant pressure reactor with the trained ANN. Figs. 3 and 4 show the time evolution of the mean mass fractions for all stochastic particles from each inlet (fuel inlet, air inlet, and burnt gas inlet) at each time step, specifically at 1 bar (case 1) and 20 bar (case 2). Although the global equivalence ratio is 0.8, combustion occurs for the particles at both leaner and richer equivalence ratios due to micromixing. Therefore, the ANN must perform well across all equivalence ratios; otherwise, errors may accumulate quickly, leading to significant deviations in the predicted trajectories compared to direct integration.



(a) Major species and T. Air inlet.

(b) Minor species. Air inlet

(c) Major species and T. H_2 inlet

(d) Minor species. H_2 inlet

(e) Major species and T. BG inlet

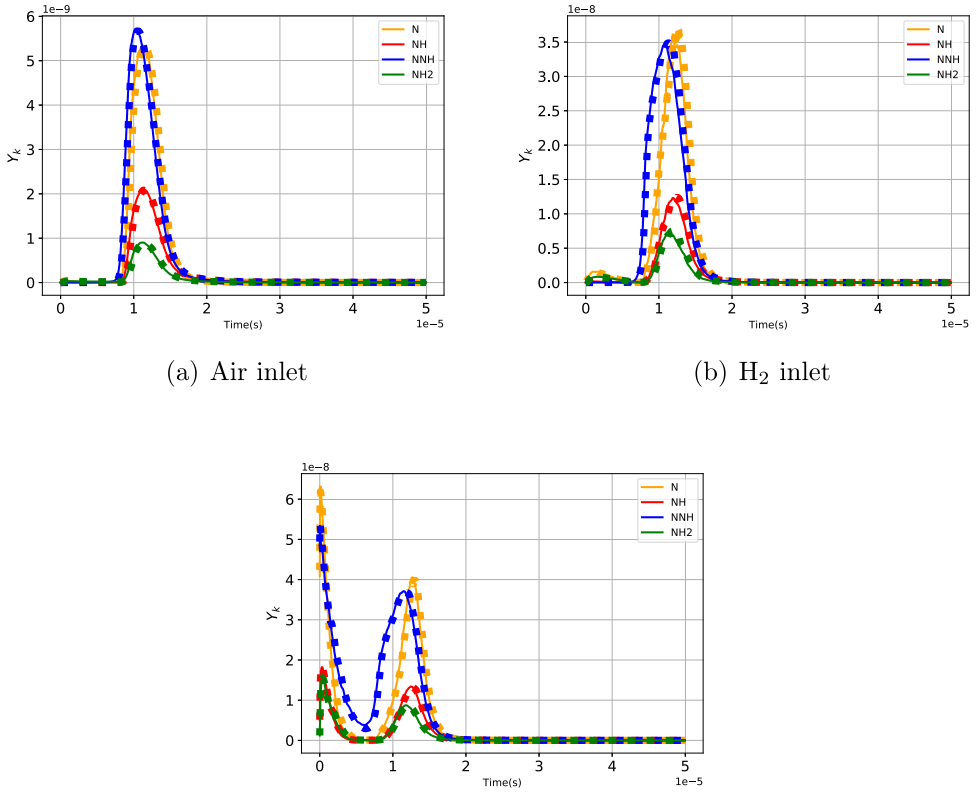
(f) Minor species. BG inlet

Fig. 11. Time-averaged evolution over particles per inlet (BG: Burnt Gases). Case 4 of Table 1, 20 bars, equivalence ratio $\phi = 0.4$. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B.

In the 1-bar case (Fig. 3), methods A and B yield similar results, while method C fails. Method C was designed for syngas combustion with multiple levels of heat losses (which are not applied in the present study) and is considered excessive for the relatively simple chemistry of hydrogen. Due to the complexity and black-box nature of the ANN, identifying the reason for the divergence in this instance is challenging. However, since the better-performing methods are not the most complex in terms of preprocessing steps, these results highlight the importance of the database in ANN training procedures. In this case, having a log-transformed and well-balanced database is sufficient to achieve a reasonable level of precision. Nonetheless, both elements are necessary; relying solely on log transformation without data augmentation results in a complete divergence of the ANN, while data addition without log transformation leads to a slight delay in ignition timing (not shown for brievty).

The 20-bar case (Fig. 4) exhibits similar trajectories to the 1-bar case but with different time scales. In this instance, all three methods show close agreement with direct integration, although method C still lags slightly behind the other two. Given that method C involves the most extensive preprocessing and training, it is ruled out. The choice between method A and method B is not as straightforward, as both require similar effort for training. Method A (random addition of radical species) is easier to implement, while method B (substepping) is more closely aligned with a physics-based approach and can be easily expanded to include additional species. Therefore, we have decided to adopt this second approach for the upcoming cases involving NOx species, based on its physics-oriented nature.

In the following results for cases 1 to 4 (Figs. 5 to 12), the 15-species mechanism is used as a reference for comparing the ANN predictions. These cases are particularly relevant for validating the proposed methodology for predicting H_2 combustion with NOx species, as they represent the most general burning conditions. Method B was selected, along with the double MLP approach described in Section 4.2. The ANN results show good agreement with the reference scheme.



(a) Air inlet (b) H_2 inlet

(c) BG inlet

Fig. 12. Time-averaged evolution of NOx over particles per inlet (BG: Burnt Gases). Case 4 of Table 1, 20 bars, equivalence ratio $\phi = 0.4$. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B.

In the results presented above, the ANNs are not constrained by any mass or atom conservation laws. The deviation in oxygen and hydrogen atoms, which should ideally be zero, reach a maximum of 0.15% in mass through one iteration. On average, across all particles, this amounts to 0.001% of the total mass for oxygen and 0.02% for hydrogen. Although these values are relatively small, this source term can accumulate, leading to significant deviations over time. At the end of the physical simulation, the maximum deviation is 2% for oxygen atoms (averaging 0.4% across all particles) and 10% for hydrogen atoms (averaging 0.6% across all particles). To mitigate this issue, a mass conservation law is now implemented to correct the predicted results.

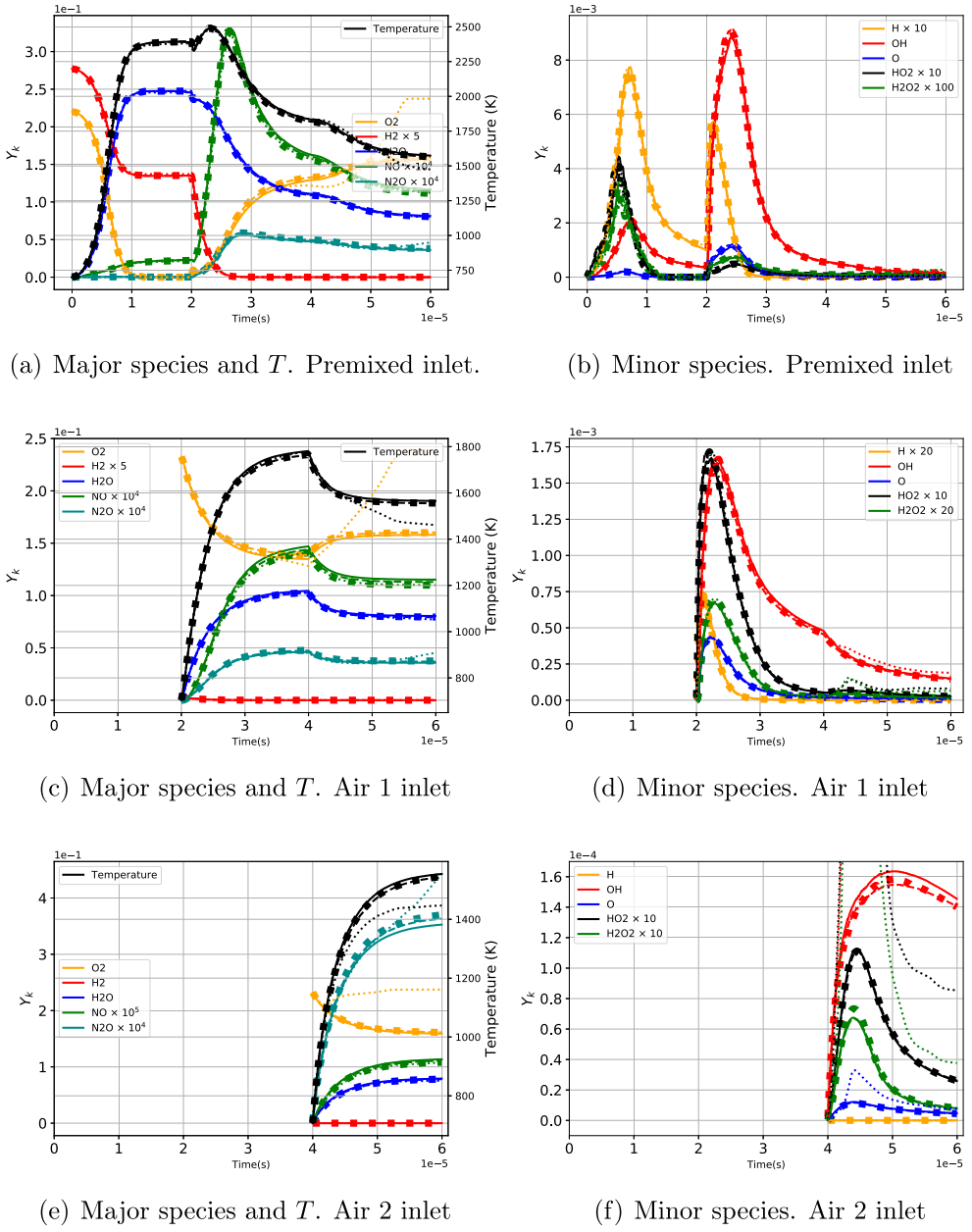
Figs. 13 and 14 present case 5 from Table 4, which simulates a more complex injection system. The same methodology was followed, and the atomic conservation algorithm proposed in [29] was implemented. A second run, including this correction, was conducted for comparison. This correction is applied after the ANN prediction, recomputing the mass fraction of one species for each atom that needs to be conserved-specifically, hydrogen and oxygen (nitrogen is already conserved since N_2 is computed from nitrogen conservation). The mass fraction to be recomputed for each particle is selected based on its greatest contribution to the oxygen and hydrogen masses, and the corrections are applied sequentially to ensure that atomic mass conservation reach machine precision. This correction generally enhances the ANN results, except for the NOx radicals (Fig. 14), and it does not significantly affect the mass fractions of other species. However, it is important to note that this adjustment may not always lead to improved outcomes, as it can compromise the ANN predictions in some cases. Interestingly, this method proves to be most effective when the ANN begins to diverge after reaching equilibrium.

In this last case, one ANN has been trained on the entire dataset (both burning stages and cooling), whereas a second one was trained with a dataset containing only the particles going through both stages of combustion (from 0 to 40 μs). The ANN trained on combustion but not on the cooling stage diverges when encountering states it was not trained on. The differences observed before 40 μs between method B and method B without dilution are mostly due to the stochastic nature of the ANN training. For example, the latter ANN returns a better predicted NH_2 mass fraction during the premixed flame stage, whereas the NNH mass fraction is better predicted by the first one. Indeed, there is always a small variability in an ANN undergoing twice identical training. This is due to the multiple local minima of the cost function created by the variability of the output data, and the stochastic nature of the mini-batch gradient descent algorithm used to reach it.

Finally, the discrepancy that can be observed on both ANNs versus CVODE integrator on Air inlet 2 (Fig. 14) is a consequence of the clipping done under $Y_k = 10^{-12}$ in order to log-transform the input data. Under this threshold, the values predicted will have a shifted behavior compared to the physical ones. However, this does not affect the other species mass fractions.

6. Discussion

Novel fuels, such as hydrogen, necessitate a thorough reevaluation of combustion systems. Among the tools used for design and optimization, numerical simulation techniques require significant adjustments and enhancements. Machine learning is increasingly integrated into these simulations and is becoming a standard tool. Replacing direct chemistry integration with ANNs has shown that strategies effective for natural gas must undergo substantial modifications to address the complexities of aeronautical combustion chambers. These chambers operate at elevated pressures, involve recirculation of burnt gases, and include secondary air-dilution.



(a) Major species and T. Premixed inlet.

(b) Minor species. Premixed inlet

(c) Major species and T. Air 1 inlet

(d) Minor species. Air 1 inlet

(e) Major species and T. Air 2 inlet

(f) Minor species. Air 2 inlet

Fig. 13. Time-averaged evolution over particles per inlet (BG: Burnt Gases). Case 5 of Table 2, 20 bars, global equivalence ratio $\phi = 0.4$. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B – – –: Method B with atoms conservation. ·····: Method B with ANN not trained on dilution stage.

The cases studied demonstrate that, for aeronautical combustion chambers, it is preferable to allocate one ANN to major species (O_2, H_2, H_2O, NO, and N_2O) while another focuses on radical species (H, OH, O, HO_2, H_2O_2, N, NH, NNH, and NH_2).

Finally, for practical applications, it is crucial to ensure that the training process covers the entire composition space expected in the combustion chamber. This includes accounting for intermediate time steps required by the stiff ODE solver to capture system dynamics and explicitly incorporating a strategy to conserve atomic mass in the ANN's prediction of species increments.

Table 6 presents the CPU time savings achieved using ANN compared to direct integration, with gains ranging from 1.5 to 1.8 depending on the specific case. These improvements are particularly noteworthy given that chemical kinetics computations in CFD can account for up to 50% of the total CPU time.

Table 6
Speed-up between direct integration and ANN on a single processor.

Case	ANN speed-up
Case 1 & Case 3 (1 bar)	1.6
Case 1 & Case 3 (1 bar) NOx	1.5
Case 2 & Case 4 (20 bar)	1.8
Case 2 & Case 4 (20 bar) NOx	1.7

7. Conclusion

In this work, a generic database was generated through a canonical micro-mixing problem to train neural networks based on two chemical schemes for H_2 combustion, both with and without NOx pollutants. The advantage of using a stochastic micro-mixing problem is its ability to cover the full range of equivalence ratios and ignition temperatures. This approach ensures that all states of the composition

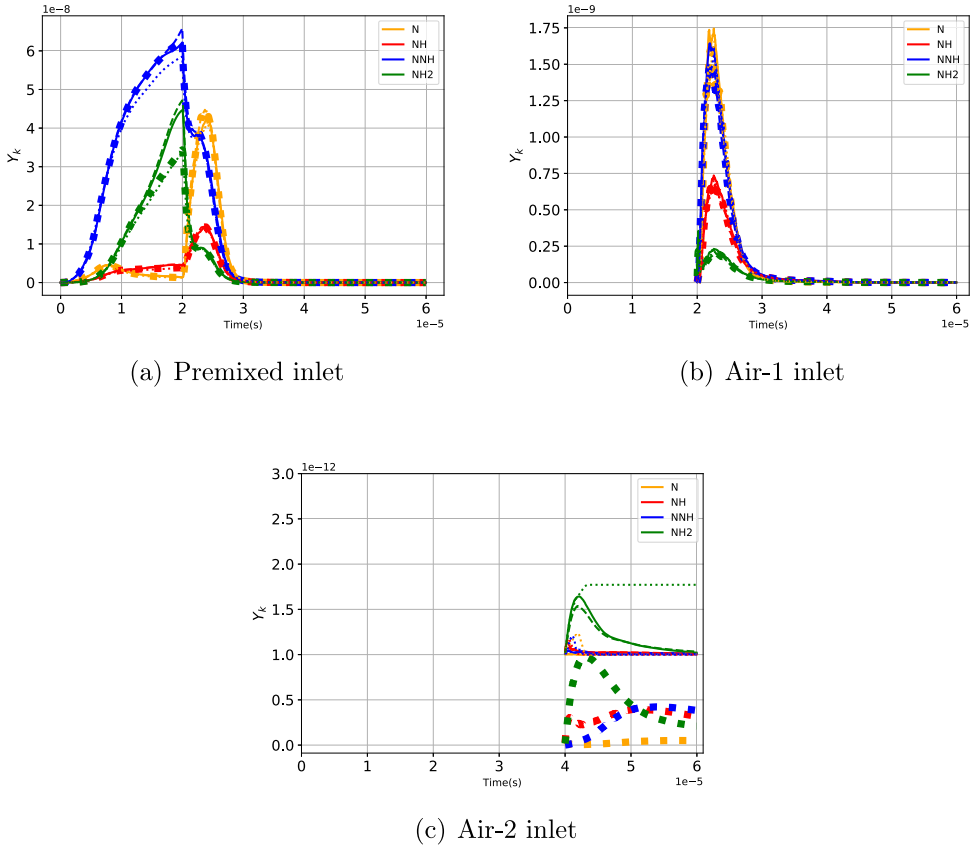


Fig. 14. Time-averaged evolution of NOx over particles per inlet (BG: Burnt Gases). Case 5 of Table 2, 20 bars, global equivalence ratio $\phi = 0.4$. ■ ■ : 15-species (H_2-NOx) mech. ——: Method B – – –: Method B with atoms conservation. ·····: Method B with ANN not trained on dilution stage.

space encountered in a turbulent reactive simulation are captured and learned by the neural networks, including various chemical time histories, thereby likely activating the numerous chemical pathways of the detailed chemistry.

Several methodologies were tested and compared, revealing that the most effective strategies involved data augmentation, which highlights the importance of the database in ANN training procedures. A methodology for predicting H_2 chemistry with NOx species was selected, based on splitting the species between two MLPs according to their expected characteristic chemical time scales. This methodology was then evaluated on four generic cases and demonstrated satisfactory performance. Finally, a case that mimicked a more realistic application was successfully tested using the same approach.

The CPU cost of reading the species mapping for the next time increment is close to two times faster than running a dedicated solver for stiff chemistry, thereby paving the way for advancements in computational fluid dynamics for reacting flows. Once the database is ready, the neural network can also be trained for time steps larger than those required for direct integration.

CRediT authorship contribution statement

Ada Béroudiaux: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Luc Vervisch:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Pascale Domingo:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ada BERODUIAUX reports financial support was provided by Safran SA. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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